

The `Boolean` class in Java is a wrapper class for the primitive data type `boolean`. It is part of the `java.lang` package and provides methods to work with boolean values, such as parsing, converting, and comparing them. The `Boolean` class also provides constants for representing the boolean values `true` and `false`.

Key Features of Boolean Wrapper Class

1. **Immutability:** Instances of the `Boolean` class are immutable.
2. **Constants:** Provides constants like `TRUE` and `FALSE`.
3. **Static Methods:** Methods for parsing strings to booleans, comparing booleans, and converting booleans to strings.

Boolean Class Methods

Here are some commonly used methods in the `Boolean` class:

1. `boolean booleanValue()`
2. `int compareTo(Boolean b)`

3. static int compare(boolean x, boolean y)
4. boolean equals(Object obj)
5. static boolean getBoolean(String name)
6. int hashCode()
7. static boolean parseBoolean(String s)
8. String toString()
9. static String toString(boolean b)
10. static Boolean valueOf(boolean b)
11. static Boolean valueOf(String s)

1. boolean booleanValue()

Returns the value of this Boolean **as a** boolean.

```
public class BooleanExample {  
    public static void main(String[]  
args) {  
        Boolean b = Boolean.TRUE;  
        boolean value =  
b.booleanValue();  
        System.out.println(value);    //  
Output: true
```

```
}  
}
```

Output:

```
true
```

2. `compareTo(Boolean b)`

Compares this Boolean object with the specified Boolean object.

```
public class BooleanExample {  
    public static void main(String[]  
args) {  
        Boolean b1 = Boolean.TRUE;  
        Boolean b2 = Boolean.FALSE;  
  
        System.out.println(b1.compareTo(b2));  
        // Output: 1  
    }  
}
```

Output:

```
1
```

3. static int compare(boolean x, boolean y)

Compares two boolean values.

```
public class BooleanExample {  
    public static void main(String[]  
args) {  
  
    System.out.println(Boolean.compare(true,  
false)); // Output: 1  
    }  
}
```

Output:

1

4. boolean equals(Object obj)

Compares this object to the specified object.

```
public class BooleanExample {  
    public static void main(String[]  
args) {  
        Boolean b1 = Boolean.TRUE;  
        Boolean b2 = Boolean.TRUE;
```

```
System.out.println(b1.equals(b2)); //  
Output: true  
    }  
}
```

Output:

```
true
```

5. static boolean getBoolean(String name)

Returns `true` if and only if the system property with the specified name exists and is equal to the string `"true"`.

```
public class BooleanExample {  
    public static void main(String[]  
args) {  
  
    System.setProperty("myBoolean",  
"true");  
        boolean b =  
Boolean.getBoolean("myBoolean");  
        System.out.println(b); //  
Output: true
```

```
}  
}
```

Output:

```
true
```

6. `int hashCode()`

Returns a hash code for this Boolean object.

```
public class BooleanExample {  
    public static void main(String[]  
args) {  
        Boolean b = Boolean.TRUE;  
  
        System.out.println(b.hashCode()); //  
Output: 1231  
    }  
}
```

Output:

```
1231
```

7. `static boolean parseBoolean(String s)`

Parses the string argument as a boolean.

```
public class BooleanExample {  
    public static void main(String[]  
args) {  
        boolean b =  
Boolean.parseBoolean("true");  
        System.out.println(b);    //  
Output: true  
    }  
}
```

Output:

```
true
```

8. String toString()

Returns a string representation of
this Boolean object.

```
public class BooleanExample {  
    public static void main(String[]  
args) {  
        Boolean b = Boolean.TRUE;
```

```
System.out.println(b.toString()); //  
Output: "true"  
    }  
}
```

Output:

```
true
```

9. static String toString(boolean b)

Returns a `String` object representing the specified boolean.

```
public class BooleanExample {  
    public static void main(String[]  
args) {  
        String s =  
Boolean.toString(true);  
        System.out.println(s); //  
Output: "true"  
    }  
}
```

Output:


```
true
```

10. static Boolean valueOf(boolean b)

Returns a `Boolean` instance representing the specified boolean value.

```
public class BooleanExample {  
    public static void main(String[]  
args) {  
        Boolean b =  
Boolean.valueOf(true);  
        System.out.println(b);    //  
Output: true  
    }  
}
```

Output:

```
true
```

11. static Boolean valueOf(String s)

Returns a `Boolean` with a value represented by the specified string.

```
public class BooleanExample {  
    public static void main(String[]  
args) {  
        Boolean b =  
Boolean.valueOf("true");  
        System.out.println(b);    //  
Output: true  
    }  
}
```

Output:

true